Equal Constraints

One Constraint

$$\sum_{i=1} a_i x_i = a_1 x_1 + a_2 x_2 + a_3 x_3 = a \rightarrow \begin{bmatrix} a_1 & a_2 & a_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} a \end{bmatrix}$$

$A\,x = b$

Three Constraints

$$\sum_{i=1} a_i x_i = a_1$$

$$\sum_{i=1} b_i x_i = a_2$$

$$\sum_{i=1} c_i x_i = a_3$$

$$\rightarrow \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

$A\,x = b$

Four Constraints

$$x_1 + 2x_2 = 1$$

$$3x_1 + 4x_2 = 12$$

$$3x_3 - 3x_1 = 10$$

$$x_1 + x_2 + x_3 = 24$$

$$\rightarrow \begin{vmatrix} 1 & 2 & 0 \\ 3 & 4 & 0 \\ -3 & 0 & 3 \\ 1 & 1 & 1 \end{vmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 24 \end{bmatrix}$$

$A\,x = b$